

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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Anonymous submission

ABSTRACT

The evaluation of generative models remains intrinsically difficult because conventional automatic metrics frequently fail to coincide with the structure of human preference. Although human judgments offer a more faithful evaluative signal, they are also expensive, time-intensive, and statistically noisy, insofar as participants differ substantially in expertise, attentional engagement, and diligence. Pairwise comparison protocols can attenuate some of this variability by eliciting relative rather than absolute judgments; nevertheless, the subsequent transformation of sparse, noisy comparisons into global quality estimates requires an explicit statistical model. Bradley-Terry-based estimators provide such a mechanism by updating item scores from observed comparison outcomes, yet existing variants either suppress rater-level heterogeneity or rely on optimization procedures without convergence guarantees, thereby limiting both robustness and interpretability. We introduce BBQ, a Bayesian Bradley-Terry formulation that explicitly parameterizes rater quality, downweights or removes unreliable participants, and obtains guaranteed monotonic likelihood improvement through an Expectation-Maximization algorithm. Across empirical evaluations, BBQ exhibits faster convergence, well-calibrated uncertainty estimates, and more robust, interpretable rankings than baseline Bradley-Terry models, including in noisy crowdsourced settings. The resulting framework supports more reliable and cost-effective human evaluation of generative models.

1 INTRODUCTION

The evaluation of generative models is especially challenging for large language models (LLMs) and image generators, where widely used automatic metrics often provide only an attenuated proxy for human preference. Measures such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, together with PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, supply limited evidence about perceived quality and therefore cannot, on their own, adjudicate fine-grained differences among systems (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Consequently, human evaluation remains indispensable for constructing meaningful inter-model rankings, which explains why leaderboards and comparative studies of both large language models and Learned Image Compression (LIC) methods increasingly rely on human preference data as a central evaluative substrate.

Human evaluation, however, is costly and slow, which makes the design of protocols that are simultaneously efficient, low-variance, and minimally susceptible to subjective noise a central methodological concern. Within this setting, pairwise comparisons, where participants choose between two candidate items rather than assigning absolute scores, constitute a practically attractive and empirically effective form of human evaluation (Zerman et al., 2018; Wang et al., 2023). Because such judgments are typically easier for participants to make, they tend to yield more consistent responses; yet exhaustive collection is infeasible, since the number of item pairs increases quadratically with the size of the evaluated set. Under the resulting sparsity constraints, naive statistics such as empirical win rates are insufficient for recovering global quality scores, because any observed win rate is intrinsically confounded by the quality of the opponents against which the item was compared.

To obtain an overall ranking suitable for leaderboards or comparative analyses, pairwise observations must therefore be aggregated through a statistically coherent procedure. This requirement mo-

054 tivate robust and computationally efficient aggregation methods, among which the Bradley-Terry
055 (BT) model (Bradley & Terry, 1952) has become a canonical approach. By iteratively estimating
056 item scores from comparison outcomes, BT models offer a principled mechanism for synthesizing
057 pairwise judgments across a broad range of domains.

058 Despite this utility, the standard Bradley-Terry model exhibits several practical limitations. BT
059 parameters can be estimated by several procedures, including iterative scaling methods (Ford Jr,
060 1957), gradient descent, and the minorization maximization (MM) algorithm (Hunter, 2004). For
061 the basic BT model, MM guarantees convergence to the global maximum, whereas gradient-based
062 procedures, which are closely related to the Elo rating system and are common in practice, do not
063 provide comparable guarantees (Hunter, 2004). Moreover, once the model is extended to include
064 factors such as home-field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce
065 et al., 1959), or ties (Rao & Kupper, 1967), even MM procedures may converge only to local optima
066 (Hunter, 2004). A distinct but equally consequential difficulty is the intrinsic noisiness of human
067 evaluation: participants may rush, guess, or lose focus, creating pronounced heterogeneity in rater
068 reliability. When such heterogeneity is left unmodeled, unreliable raters can disproportionately
069 perturb item scores and thereby diminish ranking stability.

070 We address these limitations with a Bayesian Bradley-Terry variant that jointly models item quality
071 and rater reliability. To the best of our knowledge, no prior work combines the BT framework with
072 a Bayesian formulation that explicitly represents rater quality while also yielding closed-form EM
073 updates. Whereas earlier approaches to noisy raters typically depend on gradient-based optimization
074 without convergence guarantees, our method provides stable monotonic improvement and produces
075 parameters with direct interpretability. The Bayesian formulation supplies regularization and char-
076 acterizes epistemic uncertainty, while the introduction of rater-specific quality parameters captures
077 systematic variation in participant behavior. As a result, the model can differentiate informative
078 judgments from unreliable ones, improving robustness when comparisons are obtained from noisy
079 or crowdsourced evaluators.

080 Our method integrates three complementary components. First, item skill scores are assigned
081 Gamma priors, whereas rater-specific quality parameters are endowed with Beta priors, thereby
082 preserving conjugacy where it is analytically useful. Second, the BT likelihood is embedded in a
083 mixture formulation in which a rater either follows the BT preference model or produces an unin-
084 formed random choice. Third, we derive an Expectation-Maximization (EM) algorithm with closed-
085 form updates for both item skills and rater qualities. These updates guarantee monotonic likelihood
086 improvement at every iteration and permit efficient inference even in large-scale evaluation regimes.

087 Empirically, we evaluate the approach on datasets spanning language-model assessment and image-
088 compression evaluation. The results show that the proposed model improves ranking stability and
089 top-item recovery relative to standard BT baselines, particularly under heterogeneous rater quality. It
090 also yields well-calibrated uncertainty estimates, thereby enabling meaningful confidence intervals
091 around inferred rankings. Finally, the inferred rater-quality parameters accord with observed agree-
092 ment patterns, providing interpretable diagnostics for the design and auditing of human evaluation
093 studies.

094 2 RELATED WORK

097 Human evaluation constitutes a central instrument for assessing generative models, particularly
098 when automatic metrics fail to capture perceptual fidelity or semantic adequacy. Eliciting a choice
099 between two items, rather than soliciting an absolute score for a single item, increases sensitivity to
100 subtle differences and reduces response variability (Zerman et al., 2018; Wang et al., 2023). Con-
101 sequently, pairwise comparisons have become common in evaluations of large language models,
102 image compression systems, and other generative technologies. Prominent examples include Chat-
103 bot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025), both of
104 which collect pairwise judgments and aggregate them through variants of the Bradley-Terry (BT)
105 model.

106 The Bradley-Terry model (Bradley & Terry, 1952), originally introduced for ranking competitors
107 in sports, assigns each item a positive skill parameter and expresses the probability of one item
defeating another as a ratio of their skills. Since its introduction, it has been widely adopted for

paired-comparison data in psychology, statistics, sports analytics, and machine learning. A substantial body of subsequent work has extended the BT formulation: the Plackett-Luce model, for example, generalizes pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959) by defining a distribution over permutations through products of successive BT probabilities (Luce et al., 1959). Other modifications specialize the framework to settings involving home-field advantages (Agresti, 2010), ties (Rao & Kupper, 1967), or comparisons among groups rather than individuals (Huang et al., 2006).

Maximum likelihood estimation for BT models has likewise received extensive attention. Zermelo (1929) introduced an iterative procedure for computing these estimates, a method that subsequently became widely used. Lange et al. (2000) later showed that this procedure is a special case of minorization-maximization (MM), in which surrogate functions are optimized iteratively to increase the likelihood. Building on this perspective, Hunter (2004) extended MM algorithms to generalized BT models and identified conditions under which convergence to the MLE is guaranteed. For the classical BT model, convexity ensures convergence to the global optimum (Hunter, 2004); however, for extensions involving home-field advantage, multiple comparisons, or ties, MM algorithms may guarantee only convergence to a local maximum (Hunter, 2004).

Bayesian formulations of the BT model have been developed to incorporate prior information and regularization (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) showed that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, thereby enabling Bayesian inference through Gibbs sampling. Such Bayesian BT models are attractive because they provide uncertainty estimates over item skills and can reduce overfitting in sparse comparison regimes. Our method extends this line of work by introducing a Bayesian treatment of rater quality alongside item quality, rather than treating comparison outcomes as uniformly reliable.

Crowdsourced evaluation introduces an additional layer of difficulty because participant quality can vary substantially, motivating models that explicitly represent annotator reliability (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates item quality and rater reliability through annotator-specific parameters, and demonstrated improved accuracy under noisy conditions. Their approach, however, relies on Gaussian priors and gradient-based optimization, a combination that can be difficult to optimize in high-dimensional spaces because of local optima and the absence of general convergence guarantees. By contrast, our EM-based formulation provides guaranteed monotonic likelihood improvement. A simplified Elo-based rater model developed by Google Research implements a related idea and is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and associated studies (Mentzer et al., 2020; Ballé et al., 2025).

To the best of our knowledge, no previous work combines the Bradley–Terry model with a Bayesian formulation that explicitly models rater quality while also admitting closed-form EM updates. Unlike Chen et al. (2013), whose model uses Gaussian priors and gradient-based optimization without convergence guarantees, our approach exploits conjugate priors together with an EM algorithm that guarantees monotonic likelihood improvement. Relative to the Elo-based rater model used in CLIC (CLIC, 2025), which functions as a simplified heuristic based on Elo-style updates, our method offers a principled Bayesian account with uncertainty estimates and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

Transforming noisy pairwise comparisons into a reliable item ranking is a fundamental problem in both machine learning and statistics. Consider a collection of K items repeatedly compared in pairs by R raters, with the observed comparison outcomes denoted by D . For any two items i and j in this collection, Bradley & Terry (1952) proposed the following probabilistic model:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ denotes the skill parameter associated with item $k \in \{1, 2, \dots, K\}$. Although this model provides a transparent and interpretable mechanism for inferring item scores from observed

wins and losses, it rests on the simplifying assumption that all comparisons are equally reliable. In human evaluation settings, that assumption is frequently untenable, because participants may differ in expertise, attention, or diligence, and these differences can generate inconsistent or unreliable judgments.

To represent heterogeneous rater reliability, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. Under this interpretation, rater r produces an informed Bradley-Terry-consistent judgment with probability q_r , whereas with probability $1 - q_r$ the rater selects randomly, equivalent to a fair coin flip between the two items. This construction yields the following mixture model for the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The corresponding observed-data log-likelihood is:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ is the total number of comparisons between items i and j made by rater r , and $w_{r,ij}$ counts how often that rater ranks item i above item j . Because this likelihood does not admit a closed-form maximum likelihood estimator, the optimal parameters λ and q cannot be obtained analytically. Although generic iterative methods such as gradient descent could be applied, they introduce learning-rate sensitivity and provide no convergence guarantee. The Expectation–Maximization (EM) algorithm, by contrast, avoids learning-rate tuning while guaranteeing monotonic improvement of the observed-data likelihood and convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we adopt a Thurstonian interpretation of the Bradley-Terry model (Diaconis, 1988). In this view, a comparison between items i and j is conceptualized as a race in which the two items possess random arrival times, denoted Y_i and Y_j . These arrival times are assumed to follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time is declared the winner. This latent-race construction immediately recovers the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the EM algorithm, we introduce latent variables that render the complete-data likelihood more tractable. We first define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which indicates whether the c -th comparison of items i and j by rater r is generated by the Bradley-Terry component. Using these indicators, we define $Z_{r,ij}$ as the aggregate of the minimal arrival times across the $n_{r,ij}$ comparisons made by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditional on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the latent variable $Z_{r,ij}$ is Gamma distributed:

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma latent-variable representation yields tractable EM updates while preserving the rater-specific quality structure of the model.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm iteratively computes maximum a posteriori (MAP) estimates for the model parameters, λ and q , by treating the rater-quality indicators and the unobserved Thurstonian arrival-time quantities as latent variables. Under the usual EM conditions, the procedure converges to a stationary point of the posterior distribution.

We first specify prior distributions for the parameters. The item skills λ_k receive Gamma priors, $\lambda_k \sim \Gamma(a, b)$, exploiting conjugacy with the exponential distribution, while each rater-quality parameter q_r receives a Beta prior, $q_r \sim B(\alpha, \beta)$.

The algorithm centers on repeated maximization of the expected complete-data log-posterior, conditioned on the current parameter estimates (λ^*, q^*) . It alternates between the following two steps until convergence:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, denoted by Q . This quantity is the conditional expectation of the log-posterior over both observed and latent variables, given the observed data and the parameter estimates from the preceding iteration:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} \left[\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q) \right]. \quad (9)$$

The complete-data log-likelihood, ℓ_c , decomposes into three constituent terms.

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step updates the model parameters by maximizing the Q function. By using the conditional expectations computed in the E-step, this maximization converts the original nontrivial optimization problem into closed-form updates. The central quantity shared by both updates is the posterior probability that a particular comparison from rater r originated from the Bradley-Terry component. Denoted by γ , this probability functions as a model-dependent weight for the rater's judgment, reflecting the extent to which the observed outcome agrees with the current predictions. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the current Bradley-Terry probability that item i beats item j . Thus, γ quantifies the posterior confidence that a comparison is informative rather than random under the current parameter estimates, with larger values corresponding to more trustworthy comparisons.

The updated estimate of rater quality, q_r , takes the form of a regularized weighted average. The numerator aggregates the rater's effective informative judgments, weighting each observed outcome by the posterior probability γ that it reflects a non-random Bradley-Terry comparison, and then incorporates the Beta-prior hyperparameters for regularization. The denominator normalizes by the rater's total number of comparisons and the corresponding prior mass. Consequently, the update increases a rater's quality estimate when that rater's judgments repeatedly align with the model's current predictions. The update is:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r denotes the total number of comparisons contributed by rater r .

The updated item-skill estimate, λ_i , is a ratio balancing two complementary quantities. The numerator sums the effective wins of item i across raters, again weighting each win by the corresponding posterior reliability term γ , and therefore represents the total positive evidence for the item. The denominator captures the item's aggregate comparison exposure against all other items and acts as a normalizing factor, while the Gamma-prior hyperparameters provide regularization. The resulting update is:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The derivation of the Expectation Maximization algorithm is provided in Appendix B.

4 EXPERIMENTAL RESULTS

To evaluate our Bayesian Bradley-Terry model with rater Quality (BBQ), we conduct a sequence of experiments comparing it with two baselines: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012) and a gradient-descent-based BT model that also incorporates rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT, we use the Google Research implementation (Google, 2025), which was employed by both CLIC (2025) and Ballé et al. (2025). The hyperparameter settings used throughout the experiments are reported in Section C.

We assess several Bradley-Terry (BT) variants across datasets that include human-preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with a newly collected inhomogeneous rater quality (IHQ) dataset derived from the CLIC2024 image test set (CLIC, 2025). To isolate the effect of rater quality, we partition IHQ into screened and unscreened subsets, which permits analysis of how low-quality, high-quality, and mixed comparison pools influence model behavior. Details on IHQ construction appear in Section E, while Section D and Table 2 summarize all datasets in terms of comparisons, raters, and items.

Because real datasets do not provide ground-truth rankings, we first validate the models on simulated data, where all methods recover the same ordering as the number of samples increases. For crowdsourced datasets, we approximate ground truth with the ranking obtained from the full dataset, and we assess the plausibility of this approximation by examining the top-ranked item under each model. Across all datasets except IHQ-unscreened, the three models identify the same top-ranked item. On IHQ-unscreened, however, Crowd-BT and Bayes-BT do not identify the reference image as the highest-quality item, whereas BBQ does.

A reliable aggregation method should reproduce essentially the same ordering under repeated sampling of the study. We approximate this form of stability through bootstrapping, which estimates ranking variability across resampled datasets. The bootstrap procedure is described in Section F. Because the WD and HiFiC studies used active pair-selection procedures—binary search for HiFiC and maximum-information-gain selection for WD—their bootstrap results should be interpreted with corresponding caution.

Performance is evaluated with two complementary metrics. Top-1 agreement denotes the fraction of bootstrap samples that recover the same best item as the full dataset, whereas Kendall’s Tau (τ) measures ordinal correlation between rankings (Kendall, 1938), with larger values indicating greater stability. Thus, Top-1 agreement emphasizes recovery of the leading item, while Kendall’s Tau characterizes stability of the full ranking.

4.1 PERFORMANCE ACROSS DATASETS

We compare the models across datasets in order to characterize both ranking accuracy and ranking stability. Table 1 summarizes Top-1 agreement and Kendall’s Tau. Datasets containing more comparisons per model, including MT-Bench, WD, and HiFiC, are generally easier; HUMAINE, however, departs from this pattern, indicating that model performance is also shaped by factors such as rater consistency and rater diversity, rather than by comparison count alone.

Overall, BBQ exhibits the strongest combination of stability and robustness across datasets. It identifies the top-performing item most often, sharing the best result on three datasets, and it recovers the overall ranking most accurately on five of the eight dataset conditions while ranking second on the remaining three.

The three cases in which BBQ ranks second in Kendall’s Tau are precisely those characterized by comparatively high-quality and homogeneous rater populations. On MT-Bench, all models recover the top item perfectly; in WD, the dataset collected by Ballé et al. (2025) contains only five raters, likely the paper’s authors, suggesting unusually careful evaluation; and in IHQ-screened, raters were explicitly filtered for quality. In these settings, where rater reliability is already high and

Table 1: Performance of BT-based aggregation methods across multiple datasets. Top: Top-1 agreement [%]; Bottom: Kendall’s Tau. Across the evaluated datasets, BBQ most frequently identifies the top-performing item, recovers the best overall ranking in more than half of the conditions, and ranks second in the remaining cases.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

relatively uniform, explicit modeling of rater quality yields limited marginal benefit, reducing BBQ’s advantage over simpler alternatives.

The IHQ experiments, which separately consider screened and unscreened rater subsets, reveal a clear dependence on rater heterogeneity. When low-quality raters are present, as in CLIC-all and CLIC-unscreened, BBQ substantially outperforms both Crowd-BT and Bayes-BT. All three models obtain their best Top-1 accuracy on the screened subset, underscoring the importance of rater selection; nevertheless, Crowd-BT’s performance declines on the full dataset despite its rater-quality component, likely because each crowdsourced rater contributes fewer than 40 comparisons, increasing vulnerability to noisy annotations. BBQ, by contrast, maintains strong performance without screening and exhibits only a modest Top-1 decrease, demonstrating robustness to low-quality raters.

Nevertheless, uniformly high rater quality is difficult to guarantee in practice. Large-scale crowdsourcing inevitably introduces behavioral variability, screening procedures impose additional cost, and subjective perceptual factors may affect even diligent annotators. Under these conditions, BBQ provides a principled mechanism for exploiting partially reliable raters while attenuating the influence of noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

As suggested by Table 1, datasets with more comparisons per model generally produce stronger performance. This comparison budget can be increased either by recruiting additional raters or by asking each rater to complete more comparisons.

Figure 1 displays the effect of both factors on the three BT variants. The left column reports performance as the number of raters varies, holding comparisons per rater at its maximum, whereas the right column varies comparisons per rater while holding the number of raters fixed at its maximum. Top-1 accuracy exhibits clear model-level separation, while Kendall’s τ remains comparatively similar across methods. Crowd-BT does not converge with only one or two raters, emphasizing the advantage of an EM procedure with convergence guarantees relative to gradient descent.

Crowd-BT and BBQ perform similarly when either the rater count or the per-rater comparison count is small. BBQ requires multiple raters and repeated comparisons per rater to distinguish rater qualities effectively. As either source of information increases, BBQ increasingly outperforms Bayes-BT. Conversely, although Bayes-BT tends to underperform overall, it can exceed Crowd-BT when only a few raters or limited per-rater data are available, because Crowd-BT cannot reliably estimate rater quality in such sparse settings.

4.3 RATER QUALITY

Figure 2 shows the relationship between agreement with the final ranking and predicted rater quality. The association is clearly positive: raters whose judgments more closely match the consensus rank-

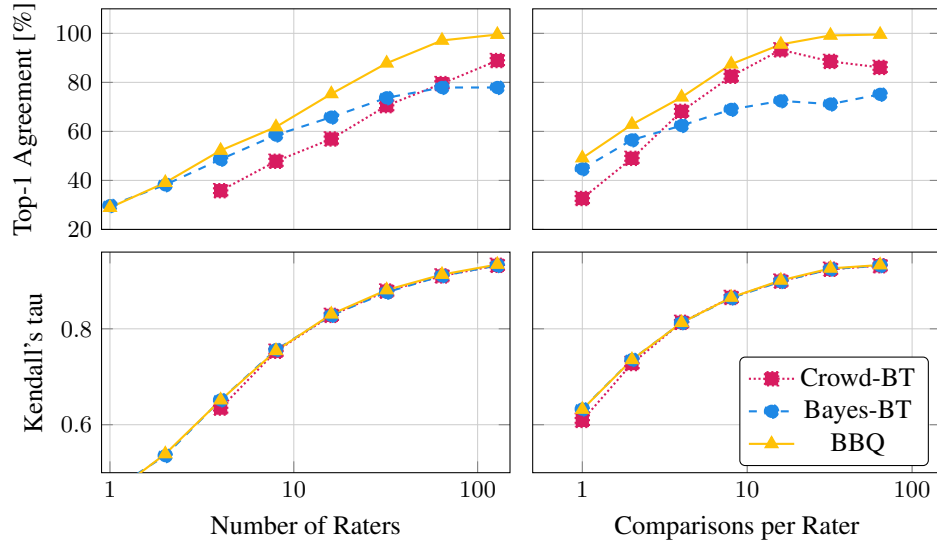


Figure 1: Scaling behavior of Bradley-Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset. **Left:** Performance as a function of the number of raters. **Right:** Performance as a function of comparisons per rater. Both Top-1 agreement and Kendall’s τ increase appreciably as additional raters or comparisons are introduced. Top-1 agreement more clearly separates the models, whereas Kendall’s τ remains comparatively similar across them. Crowd-BT fails to converge with very few raters, illustrating the practical advantage of the EM algorithm. Although Crowd-BT and BBQ behave similarly in sparse regimes, BBQ surpasses Bayes-BT as the number of raters or comparisons increases.

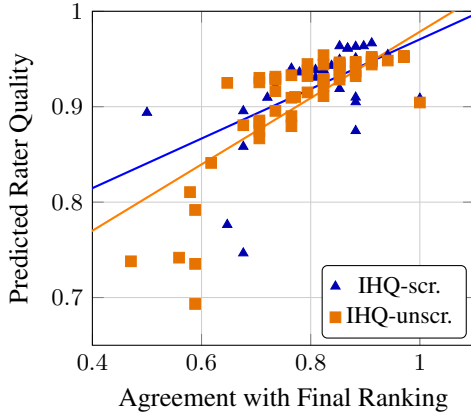


Figure 2: Scatter plot comparing rater agreement with the final ranking (x-axis) and predicted rater quality (y-axis) for the IHQ datasets. Each point corresponds to an individual rater. Triangles denote the filtered dataset, and squares denote the unfiltered dataset.

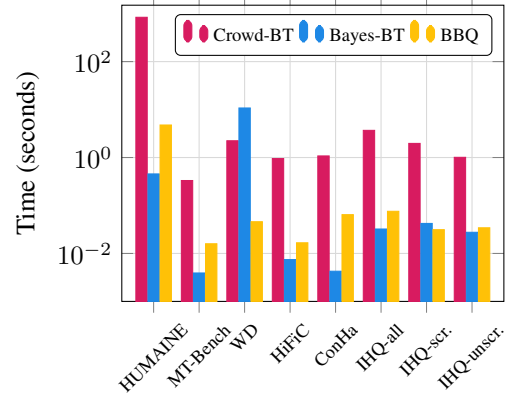


Figure 3: Average computation time in seconds, shown on a log scale, for the three models on a single bootstrap sample across eight datasets. Although Crowd-BT can require substantial computation on some datasets, BBQ remains consistently fast across all evaluated conditions.

ing receive higher model-assigned quality values. For the filtered dataset, the Pearson correlation is $r = 0.551$ across 50 raters. For the unfiltered dataset, the correlation is stronger, with $r = 0.724$ across 62 raters.

As expected, the unscreened dataset contains several raters with both lower agreement and lower predicted quality. Some raters in IHQ-unscreened even fall below the random-guessing level of

50% agreement, indicating systematic disagreement with the majority. This pattern underscores the importance of modeling rater quality when aggregating pairwise comparison data. BBQ identifies these raters and assigns them lower quality scores, thereby reducing their influence on the final ranking and mitigating the noise they introduce.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average computation time, in seconds, required for each method to process a single bootstrap sample across datasets. These timings provide a practical measure of feasibility in real-world evaluation settings. BBQ consistently converges within a few seconds on every dataset. Crowd-BT is slower than BBQ throughout, with especially long runtimes on comparison-heavy datasets such as HUMAINE, where convergence requires approximately 15 minutes. Bayes-BT converges slowly on WD, although on the other datasets it is slightly faster than BBQ. BBQ’s efficiency follows from the closed-form EM updates derived above, which enable rapid convergence even on large datasets.

These timings are particularly notable because Crowd-BT was highly optimized and implemented in C (Kernighan & Ritchie, 1988), whereas BBQ uses plain NumPy without specialized optimization. This implementation gap suggests that a compiled or more aggressively vectorized BBQ implementation could be faster still. Consequently, BBQ is not merely more robust and stable, but also practical for large-scale human-preference studies and for applications requiring repeated bootstrapping. Its combination of accuracy, stability, and computational speed makes it well suited to real-world deployment.

4.5 CONFIDENCE INTERVALS

We further examine how Crowd-BT, Bayes-BT, and BBQ quantify uncertainty in item rankings. In a controlled simulation, raters compare equally skilled items under H_0 . Each model estimates item skills and constructs uncertainty intervals for testing significant differences, as detailed in Section G. Figure 6 reports the resulting Type I error rates. At the 99% confidence level, the expected error rate is 1%. Crowd-BT and BBQ track this target closely, with slightly lower rates, whereas Bayes-BT is more conservative at $\sim 0.1\%$. Accordingly, Crowd-BT and BBQ yield well-calibrated uncertainty estimates, while Bayes-BT underestimates false positives.

5 LIMITATIONS

Although BBQ effectively models rater quality, its empirical advantages naturally diminish when all raters are uniformly reliable. In such homogeneous datasets, explicit variability modeling contributes little additional information beyond what simpler BT variants already exploit. Ensuring consistently high-quality raters, however, often requires substantial cost, screening effort, and logistical control, all of which may be infeasible in large-scale studies. This tradeoff indicates that BBQ is most valuable in realistic crowdsourced settings, whereas its incremental benefit is smaller when evaluations are carefully curated.

A second limitation concerns the class of feedback supported by the framework. BBQ is presently restricted to pairwise comparisons, whereas many human-evaluation protocols elicit ratings, full or partial rankings, or multi-way judgments. Extending the model to accommodate these feedback modalities would substantially broaden its applicability.

Methodologically, our experiments rely on a finite bootstrap procedure for estimating ranking stability. Although this procedure provides a practical variability measure, larger-scale studies would further substantiate robustness. In addition, the model assumes independence across comparisons and does not represent contextual or order effects, both of which may influence human judgments in practical evaluation interfaces.

Finally, although BBQ is computationally efficient in the evaluated regimes, extremely large item or rater populations could still require optimized, vectorized, or compiled implementations.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability. The central contribution is an Expectation-Maximization (EM) algorithm that simultaneously estimates item skills and individual rater quality, thereby downweighting or eliminating the influence of unreliable participants. By explicitly representing rater quality, the method produces rankings that are more stable and accurate under heterogeneous evaluation conditions. The EM algorithm additionally provides rapid convergence and monotonic likelihood improvement, addressing important limitations of gradient-based alternatives.

Across diverse datasets, BBQ consistently attains high Top-1 agreement and strong Kendall’s Tau, demonstrating robustness as well as ranking reliability. The model is especially effective in settings with noisy or crowdsourced raters, maintaining accuracy even when individual raters contribute relatively few comparisons. This property is particularly important for large-scale evaluations in which many raters are non-experts or vary substantially in attentiveness and diligence. The results therefore show that incorporating rater quality is most consequential when evaluation quality is heterogeneous or partially unreliable. In addition, the Bayesian framework supplies principled uncertainty estimates for item scores, enabling interpretable comparisons across studies. We further show that the resulting error bars are well calibrated and can support assessments of whether differences between items are statistically significant. Finally, predicted rater quality aligns with agreement to the final rankings, validating the model’s capacity to identify reliable evaluators.

Overall, BBQ advances human-evaluation methodology by providing a practical, interpretable, and generalizable approach for aggregating noisy pairwise comparisons. The method represents a step toward more cost-effective, statistically interpretable, and reproducible human studies for evaluating AI-generated content.

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A USAGE OF LLMs

During the preparation of this paper, we used large language models (LLMs) as auxiliary writing and development tools. ChatGPT, Claude, Gemini, and Grammarly supported spellchecking, phrase refinement, and text condensation to improve clarity and readability. ChatGPT, Claude, and Cursor were also used for code analysis, code completion, and visualization generation within the development workflow. These systems served only as supportive aids; all core research ideas, experimental design decisions, and interpretations of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider a setting in which R raters perform pairwise comparisons over K items, while each rater r is endowed with an individual quality parameter q_r : with probability q_r the observed judgment is generated by the Bradley–Terry component, whereas with probability $1 - q_r$ it is produced by random choice. The derivation below instantiates a standard EM (Dempster et al., 1977) construction in which these rater-specific reliability variables are integrated into the Bayesian estimation framework rather than treated as exogenous nuisance variation.

B.1 MODEL DEFINITION

We use the following notation, which separates item skill, rater reliability, observed comparison counts, and the latent quantities introduced for the complete-data representation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley-Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley-Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley-Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Under this notation, the Bradley-Terry comparison probability is:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

The corresponding rater-quality mixture, which interpolates between informed comparison and random selection, is:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Finally, the latent exponential representation used in the EM construction is:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

The complete-data objective is obtained by augmenting the observed comparisons with both the latent rater-quality indicators and the latent arrival-time variables, yielding the quantity:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

The contribution associated with $P(Z \mid A, \lambda)$ captures the Gamma-distributed latent arrival-time aggregates and takes the form:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

The contribution associated with $P(A \mid q)$ encodes the Bernoulli reliability mechanism for each rater:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

The comparison-outcome contribution, conditional on the Bradley-Terry indicators, is:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Combining these terms gives the complete-data log-likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: a Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and a Beta distribution $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expected complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

The posterior probability of a Bradley-Terry-consistent annotation is:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

The expected sufficient statistics are:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

The expected complete-data log-posterior is:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

The priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

The final Q -function is:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function with respect to the parameters (λ, q) while holding fixed the expectations computed in the E-step.

Update for q_r The update for each rater-quality parameter q_r is obtained by maximizing Q with respect to q_r , including the Beta-prior contribution.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i (including the Gamma prior).

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we adopt the implementation of Google (2025) without modifying its default hyperparameter configuration.

For Bayes-BT and BBQ, the hyperparameterization is limited to the prior-distribution parameters and the stopping thresholds, thereby keeping the comparison among methods relatively controlled. For both models, convergence is declared once every ELO score changes by at most 1 between successive iterations. The ELO score is obtained from the skill parameter λ through the transformation:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where ELO_SCALE_FACTOR is fixed at 400.

For the skill parameters in both Bayes-BT and BBQ, we use a Gamma prior with shape 5 and rate 0.1; for BBQ, the Beta prior on rater quality is parameterized by $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate the models on a heterogeneous collection of human-preference datasets spanning both language and image domains, with Table 2 summarizing the resulting experimental corpus.

For natural language, we use the HUMAINE dataset (Team, 2025), a large-scale benchmark containing more than 100k comparisons, together with MT-Bench (Zheng et al., 2023), which provides crowd-worker judgments comparing models on multi-turn dialogues.

For image compression, we consider three datasets: WD (Ballé et al., 2025), a densely expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which focuses on conditional generative models. Because these datasets differ substantially in scale, rater count, and rater expertise, they provide a broad testbed for assessing robustness under materially different evaluation regimes.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, obtained from a user study on CLIC2024 (CLIC, 2025) data conducted through the *Mabyduck* platform. This dataset contains two-alternative-forced-choice (2AFC) judgments over 28 generative image-compression models. To isolate the effect of rater quality, we provide two complementary subsets: (i) **screened**, in which raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, in which all raters are retained. The screened subset represents a higher-quality rater pool, whereas the unscreened subset more closely approximates the noisy participation conditions typical of large-scale human evaluations.

Table 2: Summary of the datasets used in the experiments. The IHQ dataset was collected on the Mabyduck platform and includes both 2AFC and 3AFC settings. For this dataset, screened subsets retain only raters who passed pre-screening tests for attention and display quality, whereas unscreened subsets preserve the complete rater pool.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

Taken together, these datasets permit analysis of large-scale, comparatively clean benchmarks as well as smaller and noisier regimes in which explicit rater-reliability modeling becomes methodologically consequential.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected through the Mabyduck platform (Ltd., 2025), whose interface is shown in Figure 4. Participants were asked, “Which image looks more similar to the reference image?”; ties were disallowed, and comparison pairs were sampled uniformly at random.

To study rater quality, we partitioned IHQ into screened and unscreened subsets. Screened raters were required to pass pre-tests assessing color vision, display contrast, and sensitivity to subtle differences, and they additionally completed training comparisons in which selection of the higher-quality image reinforced the intended evaluation criterion.

Figure 5 displays four representative images from the pre-screening process: the first two are standard color-blindness tests requiring correct identification of the number shown in each pattern, whereas the final two are shape-detection tests designed to evaluate sensitivity to low-contrast objects, one containing a light-gray shape on a white background and the other containing a dark-gray shape on a black background. These pre-screening tests help ensure that the screened subset contains only raters with adequate visual capabilities.

F BOOTSTRAPPING DETAILS

Because both BBQ and Crowd-BT explicitly model rater quality, we bootstrap over raters rather than over individual comparisons, thereby preserving each rater’s comparison distribution and producing a more appropriate assessment of ranking stability. For all datasets, we draw 10,000 bootstrap resamples of raters, except for HUMAINE, where a single Crowd-BT bootstrap iteration takes approximately 15 minutes, as discussed in Section 4.4; for HUMAINE, we therefore use 1,000 resamples to reduce computational cost.

G UNCERTAINTY ESTIMATION DETAILS

Beyond producing rankings, uncertainty estimation is necessary for determining whether observed differences are statistically significant.

Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) quantify uncertainty through the posterior distribution over item skills. Each skill parameter receives a Gamma prior that is updated using the observed pairwise comparisons. Credible intervals derived from the posterior are subsequently transformed to the Elo scale to facilitate item-level comparison.

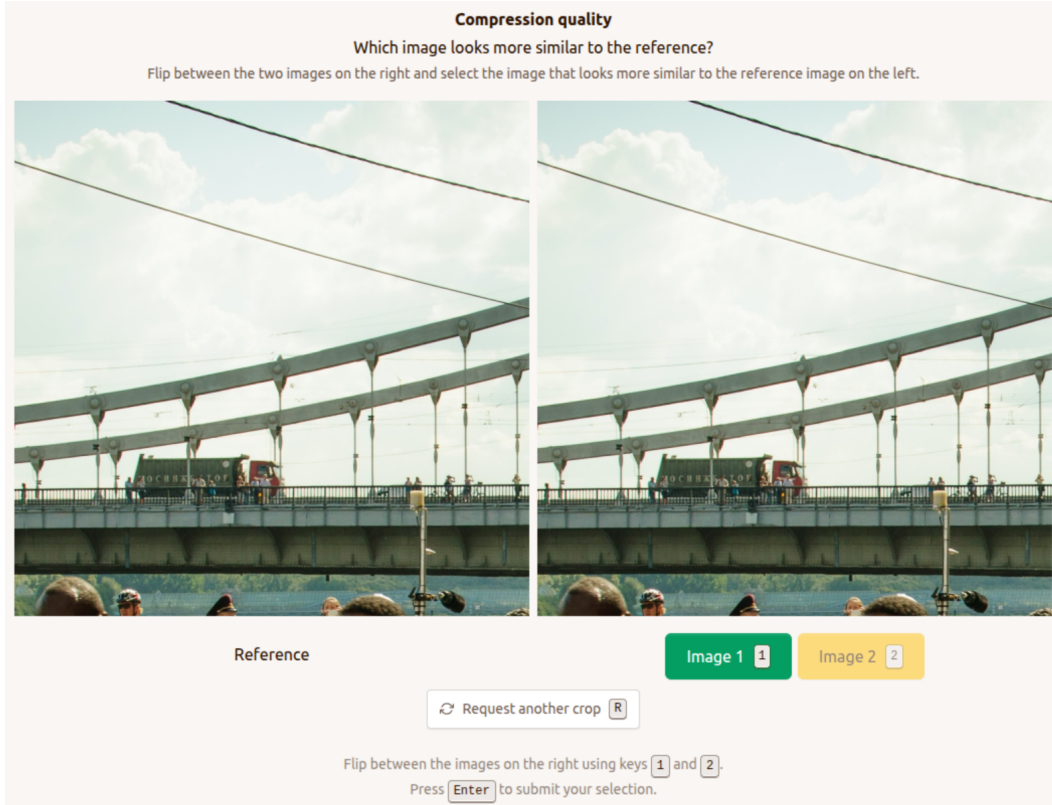


Figure 4: Screenshot of the Mabyduck user-study platform used for collecting pairwise comparisons. The reference image is presented on the left, while the rater selects the preferred alternative from two compressed images on the right.

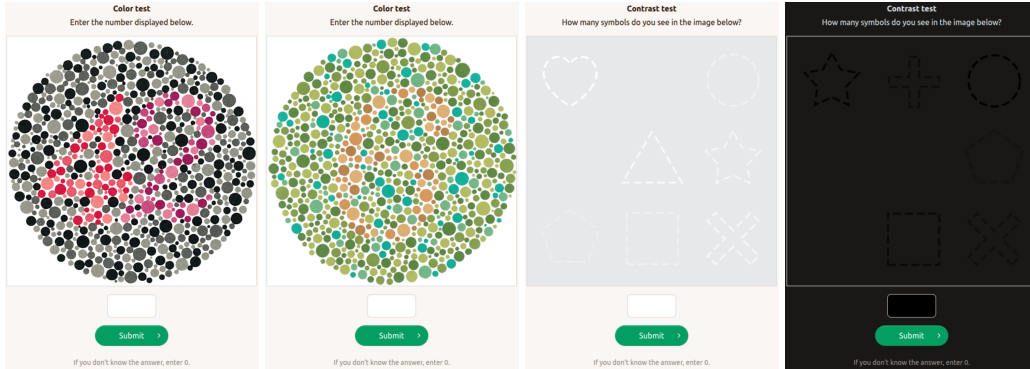


Figure 5: Four example images from the rater pre-screening procedure. The first two are color-blindness tests requiring identification of the number embedded in each pattern, whereas the final two are shape-detection tests assessing sensitivity to low-contrast objects: one light-gray shape on a white background and one dark-gray shape on a black background.

Because the classical Crowd-BT model is non-Bayesian, it estimates uncertainty using a frequentist approximation. Specifically, it applies a second-order Taylor expansion around the maximum likelihood estimate. The Hessian of the log-likelihood is inverted to obtain a covariance matrix, whose diagonal entries provide the variances of individual item estimates. These variances are converted into 99% confidence intervals as follows:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

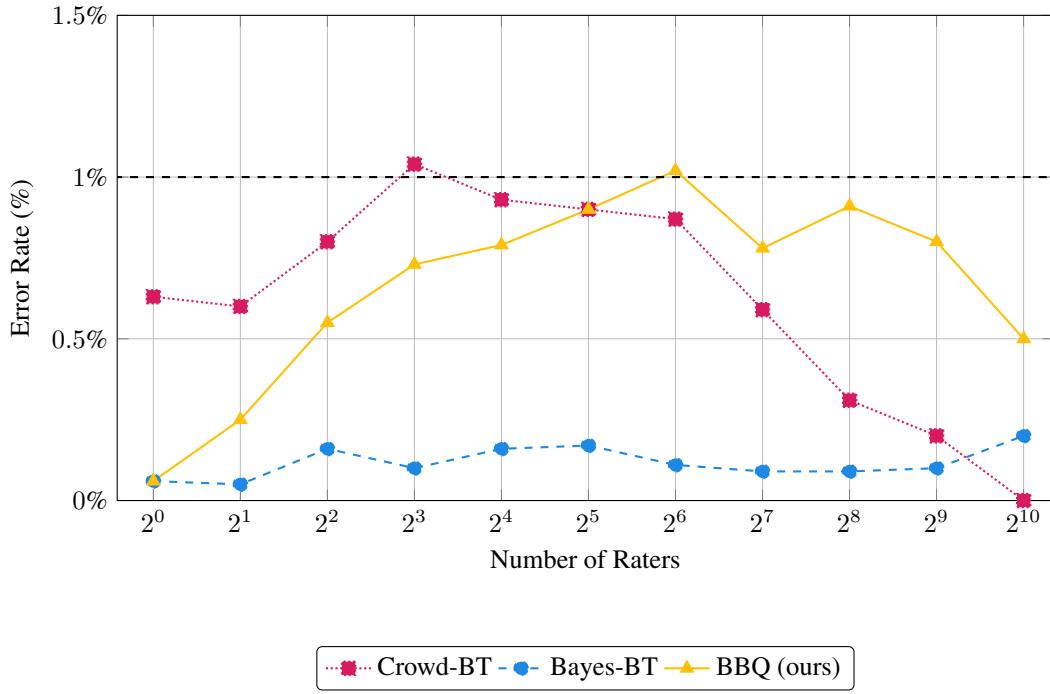


Figure 6: Type I error rate as a function of the number of raters. The dashed reference line denotes the expected 1% error rate.

where the constants rescale the standard deviation to the 99% confidence level. Sharper likelihood peaks produce narrower intervals and therefore higher apparent certainty, whereas flatter peaks produce wider intervals and indicate greater uncertainty.

To evaluate uncertainty calibration, we conduct a controlled simulation experiment. We generate trials containing two equally skilled items, corresponding to a 50–50 win probability, and simulate pairwise comparisons through coin flips. For each trial, multiple raters are simulated, after which the models estimate Elo scores together with associated uncertainty.

Formally, the null hypothesis H_0 states that the two items have equal strength. For each trial, we compute the 99% confidence or credible interval for each item’s skill estimate. A model incorrectly rejects H_0 when the intervals do not overlap, thereby indicating a statistically significant item difference despite equality in the data-generating process. The primary metric is the frequency with which each model reaches this erroneous conclusion, namely the observed Type I error rate.

We run 10,000 trials with 50 comparisons per rater while varying the number of raters, thereby assessing how uncertainty estimates scale with data volume. This design permits estimation of the empirical Type I error rate for each model and comparison with the theoretical 1% expectation at the 99% confidence level. As shown in Figure 6, Crowd-BT and BBQ produce Type I error rates close to the expected 1%, indicating well-calibrated uncertainty estimates. Bayes-BT, in contrast, is overly conservative and consistently yields lower-than-expected error rates, suggesting that its credible intervals are less well calibrated for significance testing.